REPORT 2 - NOSE CONE ANALYSIS FOR 650MM LENGTH

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Weisshorn – Nose Cone at 650 mm, with and without the 3U

EPFL ROCKET TEAM

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Summary

Weisshorn nose cone at 650 mm (curved part + coupler cylinder): two studies		
	A: 3U in the nose (as required)	**B: 3U not in the nose**
Profile	LV-Haack (Haack $C = \frac{1}{3}$)	Von Kármán (Haack $C = 0$)
Length	561 mm curved + 89 mm coupler	630 mm curved + 20 mm coupler
Apogee (model)	8703 m (–297 m vs. 9 km)	8905 m (–95 m)
9 km needs	830 mm total (LV 5.51 m)	750 mm total (LV 5.43 m)
Both	180 mm diameter; spherical tip $r_n = 4.5$ mm; CanSat in the nose	

The structure requirements cap the nose at 650 mm, curved part and cylindrical part (coupler) together, and put the 3U payload ($100 \times 100 \times 300$ mm, the guided recovery vehicle) in the nose; SE specified that if the 3U does not fit, the nose is extended with a cylindrical part. This report redoes the design of report 1 (810 mm Von Kármán) under these limits for the three best profiles of report 1, with the 3U in the nose (study A) and without it (study B), and gives the apogee of a longer nose for the interface meeting. It covers the FD scope (profile, lengths, drag, apogee, stability); joint loads, heating and the mass budget are left to ST.

- **A: the 3U decides the profile.** The 3U needs a 77 mm radius, which a slender nose reaches only near its base, so the cylinder has to grow, and every millimetre of it is taken from the curved part. LV-Haack is the fullest near the base and needs the shortest cylinder (89 mm): it flies 56 m higher than Von Kármán (155 mm cylinder) and 83 m higher than power $\frac{1}{2}$ (186 mm).
- **B: the shortest cylinder and the lowest drag.** Without the 3U the cylinder is only the 20 mm coupler overlap and Von Kármán is the lowest-drag profile (power $\frac{1}{2}$ is within 7 m, but meets the cylinder at an angle). With this nose the 3U would reach 93 mm past the joint, more than the 50 mm free in the RE tube, so it has to go elsewhere in the vehicle.
- **For the interface meeting.** Taking the 3U out of the nose is worth 201 m at 650 mm, as much as 1.4 kg of dry mass (143 m/kg). A longer nose is worth 2.7 m/mm in A and 1.1 m/mm in B near 650 mm. With the rest of the LVO unchanged the 5.4 m LV cap (SE_REQ_04) is reached at a 720 mm nose, which gives 171 m in A and 71 m in B, 58 % and 75 % of the shortfall to 9 km. Neither study reaches 9 km within the cap (Table 4), so neither gain is negligible.
- **New CFD and CAD.** 24 Euler runs at $L/D = 2.5$ and 3.5 confirm the report-1 interpolation (within 0.8 %) and correct the extrapolation below $L/D = 3$ (6–10 % too much wave drag). STEP, STL and profile CSV of the six design points (three profiles, two studies) are in `CFD/geometry_650/` for the Fluent runs.

1 Requirements and Inputs

Report 1 (2027_C_FD_REPORT-1_NOSE-CONE) chose the profile for a nose of free length and recommended a Von Kármán nose of 810 mm. The structure (ST) requirements of Weisshorn now cap the nose at 650 mm in total, and systems engineering (SE) specified that this length is the curved part plus the cylindrical part (the coupler), and that the nose is extended with a cylindrical part if the 3U payload does not fit in the curved shape. Table 1 lists the requirements that drive the nose; everything else (vehicle, engine, trajectory model, Euler solver) is unchanged from report 1.

Table 1: Requirements that drive the nose (sheet 2027_C_SE_REQ unless stated).

Requirement	Content
ST Nosecone/Payload-Bay Req_03	Total length $\leq$ 650 mm: curved part + cylindrical part (coupler), confirmed by SE
Req_13, Req_01	Payload bay prepared for both cases: a CanSat at the front of the nose with a volume for the guided recovery in its lower part, or a 3U volume; if the 3U does not fit, the nose is extended with a cylinder (SE)
PL_GRV_REQ_02, _03	3U payload (guided recovery vehicle, GRV) $100 \times 100 \times 300$ mm, 3.0 kg to 3.99 kg
ST_REQ_09	Integration diameter through couplers $\geq$ 155 mm
RE_REQ_08	Recovery (RE) assembly $\leq$ 550 mm in a 600 mm tube: 50 mm free
SE_REQ_01, _04	Apogee $\geq$ 9000 m AGL; launch vehicle (LV) shorter than 5.4 m

The coupler. As on Firehorn, the coupler ring is bonded inside a full-diameter cylindrical extension of the nose shell, behind the curved part; its aft face is the joint (the separation plane) to the RE bay. This cylinder is part of the outer skin, so its length c counts in the 650 mm and is taken from the curved part: $L = 650\,\text{mm} - c$. Its shortest length is set by ST (bond overlap, separation mechanism, fasteners). The study assumes $c \geq 20$ mm; a longer cylinder is simply a payload bay.

The 3U envelope. The 3U (100×100 mm section) needs an outer radius $r \geq 77$ mm: 70.7 mm half-diagonal, 3 mm clearance, 2 mm guide rail and the 1.2 mm shell. This is true only near the base of a slender nose. Its length is taken as 300 mm (PL_GRV_REQ_02) plus 10 mm axial clearance; report 1 used the 340.5 mm of a 3U CubeSat, whose effect is given in Section 3.5. The coupler ring leaves at least 155 mm free, enough for the 151 mm the 3U needs, so the 3U can pass through it. The CanSat (Ø66 mm$\times$115 mm) needs $r \geq 39.2$ mm by the same rule.

The two studies.

- **Study A, 3U in the nose** (the requirements as written): the cylinder is made as long as needed for nothing of the 3U to reach past the joint, within the 650 mm.
- **Study B, 3U not in the nose**: the cylinder is the 20 mm minimum and the 3U goes elsewhere in the vehicle. The nose keeps the CanSat.

For the interface meeting both studies are repeated for allowed total lengths from 600 to 850 mm. An extension of the cylinder beyond 650 mm (the other reading of SE's rule) is a point of these curves.

Unchanged from report 1. LVO vehicle (75 kg wet, 4.5 kN for 9.1 s, 180 mm), assumed fins, point-mass trajectory from the 12 m rail at $84°$, ISA, and the drag factor k that makes the LVS vehicle (Von Kármán, 720 mm) reach 9000 m. The apogee shortfalls below are therefore relative to what the LVO assumed; the real margin depends on the LVO's own margin above 9 km. The vehicle behind the nose joint (4.68 m) is the same for every option: with a 650 mm nose the LV is 5.33 m long, and a longer nose lengthens it by the same amount.

2 Method

The chain of report 1 is reused: the Euler wave drag of each nose feeds a drag build-up and a point-mass ascent trajectory, which gives the apogee. Three things are new: direct Euler runs at the short lengths the 650 mm limit allows, the fixed-total-length geometry, and the 3U fit. The study is run by `studies/total_length.py` with `configs/weisshorn_650.yaml` in the `weisshorn-nose-cone` repository.

Profiles. The three best profiles of report 1 are compared. Von Kármán (Haack series $C = 0$) has the lowest wave drag for a given length. LV-Haack ($C = \frac{1}{3}$) is the Haack nose of lowest drag for a given length and volume: it is fuller near the base, which is where the 3U sits. The power-$\frac{1}{2}$ nose ($y = R\sqrt{x/L}$) has almost the same drag as Von Kármán at short lengths but is the slimmest at the base. All three get the 4.5 mm spherical tip of report 1.

Wave drag. Report 1 computed these profiles at $L/D = 3$, 4, 4.5 and 5 and interpolated $C_{D,\mathrm{w}}(L/D)^2$, which slender-body theory predicts to be nearly constant, between them. The 650 mm limit gives $L/D = 2.5$–3.5, at and below the edge of that data. The same axisymmetric Euler solver (HLLC, MUSCL, 160×80 cells; validated against Taylor–Maccoll to 0.5 % and grid-converged to 0.5–1 % in report 1) was therefore run directly at $L/D = 2.5$ and 3.5 for the three profiles at $M = 1.6$, 1.4, 1.2 and 1.1, giving 24 new points (study `f650`). The transonic rise below $M = 1.1$ is modelled as in report 1.

Fixed total length. For a coupler of length c, the curved part is $L = 650\,\mathrm{mm} - c$. The vehicle behind the joint is fixed, so the coupler cylinder adds its wetted area $\pi D c$ to the skin friction and its shell mass (0.98 kg/m^2, the Firehorn as-built areal density) to the vehicle; the curved shell mass and the tip follow report 1.

3U fit. With the profile $y(x)$ measured from the apex, the 3U fits in the last l_{in} of the curved part, where $y \geq r_{\mathrm{req}} = 77$ mm:

$$l_{\mathrm{in}} = L - x^*, \qquad y(x^*) = r_{\mathrm{req}}. \tag{1}$$

For all three profiles y/R depends only on x/L, so $l_{\mathrm{in}} = \kappa L$ with a constant κ: 0.31 for Von Kármán, 0.39 for LV-Haack and 0.27 for power $\frac{1}{2}$. Placed as far forward as possible, the 3U of length $l_{3\mathrm{U}}$ (with its clearance) then occupies the coupler and reaches

$$p = l_{3\mathrm{U}} - l_{\mathrm{in}} - c = l_{3\mathrm{U}} - \kappa\,(650\,\mathrm{mm}) - (1-\kappa)\,c \tag{2}$$

past the joint. Every millimetre of coupler pulls the 3U back by only $1 - \kappa$ of a millimetre, because it also shortens the curved part. The design point of a profile is, in study A, the shortest coupler (≥ 20 mm) with $p \leq 0$ and, in study B, the 20 mm minimum. For the interface meeting the same rules are applied with the 650 mm replaced by an allowed total length from 600 to 850 mm; the vehicle behind the joint is kept, so the LV grows with the nose.

Stability. As in report 1: the static-margin change uses Barrowman's nose centre of pressure $x_{cp} = L - V/A_{\mathrm{ref}}$ (V the nose volume) and the nose mass, relative to the LVS nose.

3 Results

3.1 Wave drag at short lengths

Table 2 gives the new direct Euler values. At $L/D = 3.5$ the report-1 interpolation between $L/D = 3$ and 4 is within 0.6 % at $M = 1.4$ and 0.8 % at $M = 1.2$–1.6, so the numbers quoted from it earlier hold. Below $L/D = 3$ report 1 had no data and held $C_{D,\mathrm{w}}(L/D)^2$ constant; the new runs show that this product keeps falling as the nose gets blunter, so the extrapolation overestimated the wave drag of the shortest noses by 6–10 %. All results below use the new data. The $M = 1.1$ runs, and the $M = 1.2$ runs of LV-Haack and power $\frac{1}{2}$ at $L/D = 2.5$, stopped at the 6000-iteration limit with a residual uncertainty of 1–2 %, as the $M = 1.1$ runs of report 1.

Table 2: Direct Euler wave drag $C_{D,\mathrm{w}}$ (sharp apex) and the error of the report-1 model (interpolation at $L/D = 3.5$, extrapolation from $L/D = 3$ at 2.5).

Profile	L/D	$M = 1.1$	$M = 1.2$	$M = 1.4$	$M = 1.6$	report-1 model at $M = 1.4$
Von Kármán	2.5	0.1156	0.1212	0.1190	0.1155	+5.7 %
Von Kármán	3.5	0.0680	0.0686	0.0667	0.0648	−0.4 %
LV-Haack	2.5	0.1147	0.1302	0.1356	0.1347	+7.1 %
LV-Haack	3.5	0.0714	0.0767	0.0776	0.0766	−0.6 %
Power $\frac{1}{2}$	2.5	0.1052	0.1100	0.1122	0.1119	+9.8 %
Power $\frac{1}{2}$	3.5	0.0641	0.0665	0.0677	0.0675	−0.5 %

3.2 Coupler length at 650 mm

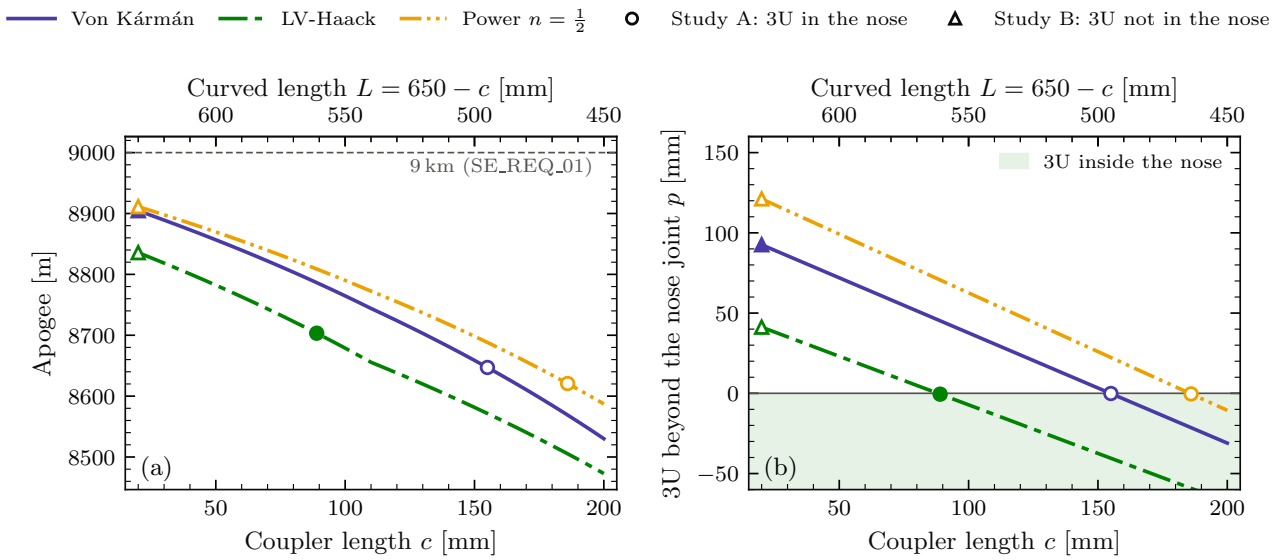


Figure 1: Total length 650 mm. (a) Apogee and (b) length of 3U past the nose joint against the coupler length c. Markers: design point of each profile in study A (circles) and B (triangles); filled: the design chosen in each study.

Figure 1a shows the cost of the coupler: the apogee falls by 1.3 m to 1.7 m per millimetre of coupler at $c = 20$ mm and by about 2 m per millimetre at $c = 100$ mm to 150 mm, because the curved part gets shorter and blunter. At equal length Von Kármán and power $\frac{1}{2}$ fly within 7 m of

each other (the difference is inside the model uncertainty), and 69 m higher than LV-Haack, as in report 1.

Figure 1b shows why this ranking does not decide study A. With the shortest coupler the 3U reaches 93 mm (Von Kármán) and 121 mm (power $\frac{1}{2}$) past the joint, and each millimetre of coupler pulls it back by only $1 - \kappa$ of a millimetre (Equation 2). LV-Haack has the largest κ, so its curve starts lowest and reaches zero first.

3.3 Design points

Table 3: Design point of each profile in each study at 650 mm. "3U in L": length of the curved part with $r \geq 77$ mm; p: 3U past the joint (in study B, where it would reach if it were in the nose). Bold: best, or within 10 m of the best.

Study	Profile	c [mm]	L [mm]	L/D	3U in L [mm]	p [mm]	Apogee [m]	vs 9 km [m]
A: 3U in the nose	Von Kármán	155	495	2.75	155	0	8647	−353
	LV-Haack	89	561	3.12	222	−1	**8703**	**−297**
	Power $\frac{1}{2}$	186	464	2.58	124	0	8621	−379
B: 3U not in the nose	**Von Kármán**	20	630	3.50	197	93	**8905**	**−95**
	LV-Haack	20	630	3.50	249	41	8836	−164
	Power $\frac{1}{2}$	20	630	3.50	169	121	**8911**	**−89**

- **Study A, 3U in the nose**: LV-Haack, 561 mm curved + 89 mm coupler, –297 m. Von Kármán needs a 155 mm coupler, which leaves it only $L/D = 2.75$, and loses 56 m; power $\frac{1}{2}$ loses 83 m. If the cylinder may instead be extended beyond 650 mm, keeping a 630 mm curved part is not the best use of the extra length: Von Kármán 630 + 113 mm (743 mm) reaches 8870 m, LV-Haack at 740 mm in total 8906 m (Section 3.4).
- **Study B, 3U not in the nose**: Von Kármán, 630 mm curved + 20 mm coupler, –95 m; power $\frac{1}{2}$ (–89 m) is equal within the model uncertainty, but meets the cylinder at an angle and is blunt at the tip. The 3U would reach 93 mm past the joint, more than the 50 mm free in the RE tube (RE_REQ_08), so it has to go elsewhere. Moving it 100 mm aft moves the burn-out centre of gravity aft by about 0.04 cal, which SE and FD have to include in the stability budget.
- **A against B**: 201 m at 650 mm, the apogee of 1.4 kg of dry mass.

3.4 What a longer nose buys

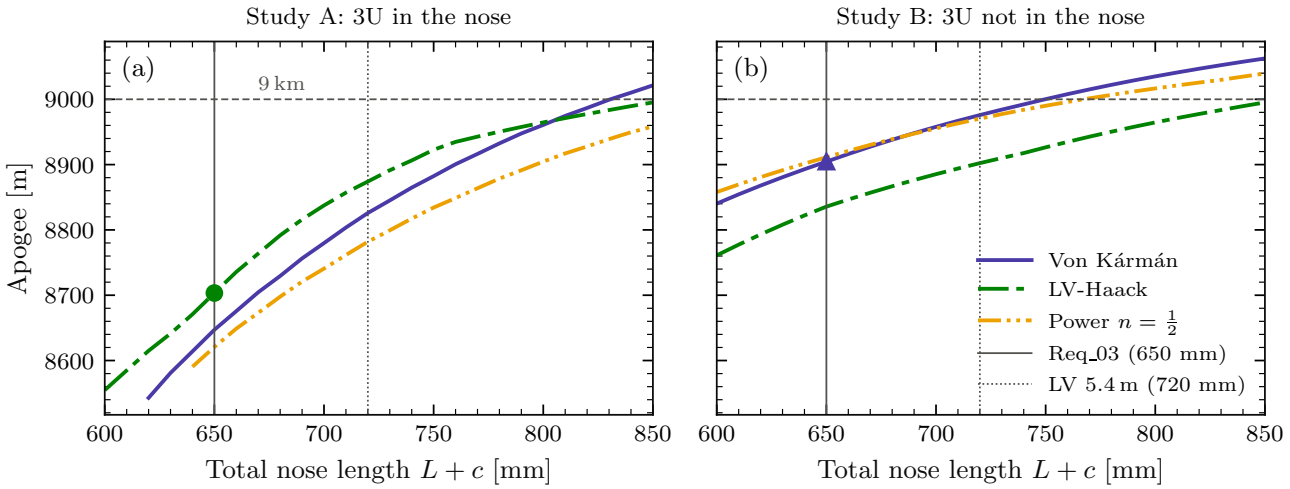


Figure 2: Apogee against the allowed total length of the nose, each profile with the coupler of its study. Solid vertical line: Req_03; dotted: LV length 5.4 m (SE_REQ_04) with the rest of the LVO unchanged. Markers: the designs of Table 5.

Table 4: Best design at each allowed total length for the interface meeting (VK Von Kármán, LVH LV-Haack; $L+c$ in mm; within 10 m, Von Kármán is listed). Gain against the same study at 650 mm. LV: launch vehicle length with the rest of the LVO unchanged (italic: not shorter than the 5.4 m of SE_REQ_04).

Total	LV	Study A: 3U in the nose			Study B: 3U not in the nose		
$L + c$ [mm]	[m]	Design $L + c$	Apogee [m]	Gain [m]	Design $L + c$	Apogee [m]	Gain [m]
650	5.33	LVH 561 + 89	8703	–	VK 630 + 20	8905	–
700	5.38	LVH 644 + 56	8837	+134	VK 680 + 20	8958	+53
720	*5.40*	LVH 677 + 43	8874	+171	VK 700 + 20	8976	+71
750	*5.43*	LVH 727 + 23	8923	+219	VK 730 + 20	9000	+96
800	*5.48*	VK 713 + 87	8961	+257	VK 780 + 20	9035	+131
850	*5.53*	VK 786 + 64	9021	+318	VK 830 + 20	9063	+158
9 km reached at		$L + c \approx$ 830 mm (LV 5.51 m)			$L + c \approx$ 750 mm (LV 5.43 m)		

Near 650 mm each millimetre of allowed length is worth 2.7 m/mm in study A and 1.1 m/mm in study B (Figure 2, Table 4). In A the extra length goes into the curved part, which also makes room for the 3U and lets the cylinder shrink: that is why the gain is larger. LV-Haack stays the best choice in A up to about 800 mm; beyond that the 3U fits in a long Von Kármán nose, which then wins. 9 km needs about 830 mm in A and 750 mm in B, both beyond the 720 mm at which the LV reaches 5.4 m. Within that cap the nose is worth 171 m (A) and 71 m (B); the rest of the shortfall has to come from the vehicle (mass, drag, engine) or from a longer LV.

3.5 Sensitivity

- **Payload length** (study A). With a 340.5 mm 3U CubeSat instead of the 300 mm GRV, the coupler becomes 156 mm with LV-Haack (–431 m), 214 mm with Von Kármán (–520 m) and 241 mm with power $\frac{1}{2}$ (–561 m, extrapolated below $L/D = 2.5$). The ranking is unchanged; every 10 mm of payload costs 33 m with LV-Haack and 41 m to 45 m with the other two.

- **Model.** A 1 % error on the wave drag moves the apogee of these noses by 5 m (study B) to 7 m (study A). The 56 m margin of LV-Haack in study A would therefore need an error of about 9 % between two profiles, against the 0.5–1 % accuracy of the solver; it comes from the geometry of the 3U fit, not from small drag differences. The absolute apogee depends on the drag factor k, calibrated on the LVO, exactly as in report 1; the differences between the studies and lengths do not.

4 Design Definition

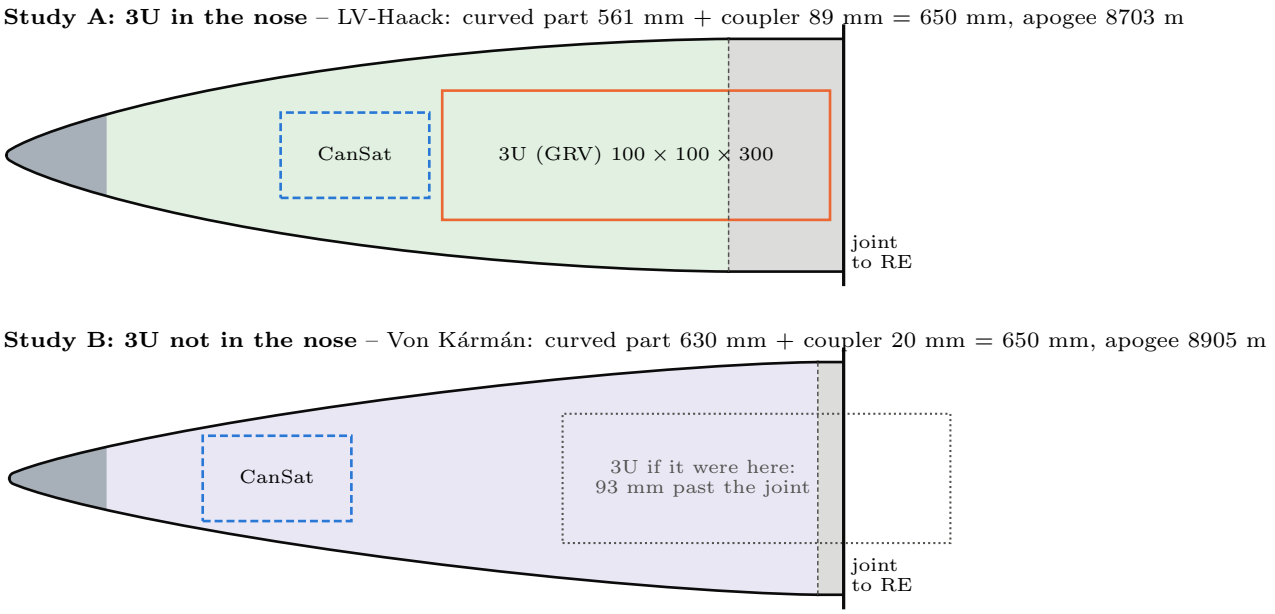


Figure 3: The two designs to scale (mm). Grey: tip and coupler cylinder; thick line: joint to the RE bay; orange: 3U; blue dashed: CanSat; grey dotted (B): where the 3U would sit if it were in the nose.

Table 5: The two nose cones. Positions from the physical tip.

	A: 3U in the nose	**B: 3U not in the nose**
Profile	LV-Haack, Haack series $C = \frac{1}{3}$	Von Kármán, Haack series $C = 0$
Haack series: $y = \frac{R}{\sqrt{\pi}}\sqrt{\theta - \frac{1}{2}\sin 2\theta + C\sin^3\theta}$, $\theta = \arccos\left(1 - \frac{2x}{L}\right)$, x from the apex, $R = 90$ mm		
Curved length L (to the theoretical apex)	561 mm ($L/D = 3.12$); 558.7 mm physical	630 mm ($L/D = 3.50$); 625.4 mm physical
Coupler cylinder	89 mm, full diameter, tangent	20 mm, full diameter, tangent
Total length	650 mm (647.7 mm physical)	650 mm (645.4 mm physical)
3U	front face at 337 mm; 222 mm in the curved part, 88 mm in the cylinder	elsewhere in the vehicle
CanSat (needs 115 mm)	220 mm of length with $r \geq 39.2$ mm ahead of the 3U	496 mm of length with $r \geq 39.2$ mm
Nose CP (Barrowman)	243 mm from the tip	310 mm from the tip
Apogee (model)	8703 m (–297 m); $M_{\max} = 1.46$	8905 m (–95 m); $M_{\max} = 1.47$
Static margin change	-0.049 cal vs. the LVS nose	$+0.018$ cal (nose only; without the 3U shift)
Both	Tip: sphere $r_n = 4.5$ mm. Diameter 180 mm.	
Geometry files	`nose_lvhaack_L561_c89_rn4.5mm`	`nose_vonkarman_L630_c20_rn4.5mm`

4.1 CAD for the Fluent runs

The six design points of Table 3 (three profiles, two studies) are exported as solids of revolution in mm, x from the physical tip, made of the blunted profile, the coupler cylinder and a flat base at the joint: STEP (OpenCASCADE B-spline through 200 points), watertight STL and the profile CSV. The STEP and CSV files are in `CFD/geometry_650/` of the repository (STL in `output/weisshorn_650/total_length/cad/`); the STEP volumes agree with the STL within 0.1 %. They replace the 810 mm geometry in the Fluent workflow note of the `CFD` folder; the set-up, Mach numbers and meshing are unchanged.